Mathematical overview: facts, assumptions and limits

An annulus has two boundary circles; a Möbius band has one. After cutting all longitudinal boundaries of n equal-width lanes, connected pieces correspond to cycles of the end-seam permutation on the lanes. Matching seam M fixes every lane: n annuli, 2n boundaries. Reversing seam R pairs lane j with n+1-j. Each two-lane cycle forms an annulus (two boundaries); with odd n the fixed middle lane remains a Möbius band (one boundary). Thus three reversed lanes yield two pieces with 2+1 boundaries; four reversed lanes yield two annuli with 2+2. Multiple visible loops may be linked in space; linking is outside these component/boundary claims.

For strips joined in a cycle, each R seam reverses the transverse direction and each M seam preserves it. The composite seam is reversing exactly when the number of R seams is odd. The resulting connected band has one boundary if odd, two if even. The number of constituent strips alone does not determine this. This is about connected bands made from full short-end joins of equal-width strips; their lengths may differ; no branching joins or extra holes occur in P2.

Removing the interior of a small embedded disk whose closure lies strictly inside a connected band, retaining the cut rim as an edge, preserves connectivity and adds exactly one boundary circle. A small notch meeting only one old boundary deforms that circle and adds none. Two corresponding half-disks at a joined seam are one disk in the assembled surface, not two. Holes/notches must remain separated, have closure disjoint from unwanted boundaries, and not span the strip width. Narrow paper bridges and tape can fail physically; they are not resolved by the topological theorem.

Children discover lane cycles, reversal parity, and new boundaries. Grades 2–3 can enact lane connections and repeated seams; Grades 4–5 can explain arbitrary lane counts and composition. K–1 may use P3's physically prepared bands and trace closed edges with adult reading. A count from one physical attempt is an observation; seam/edge arguments explain why correctly constructed models have those counts.

Materials, preparation and entry route

Per pair: four 300×80 mm strips for P1 (3M,3R,4M,4R); three 250×60 mm reusable strips with removable seam labels for P2; four 300×60 mm strips for A–D in P3; two spare 300×60 mm strips for designs; paper tape, pencil/colored markers, twelve removable edge dots, one tray. Total 13 strips. For the four-strip P2 designs, borrow one uncut 300×60 mm spare and join it to the three 250×60 mm strips; only widths must match. Dismantle and rejoin the same four strips for the second design, then return the uncut spare before P3. Mark lane cuts at 80/3 and 160/3 mm for three lanes, 20/40/60 mm for four. Label lane centers at each end and mark upper/lower corners with distinct dot/square symbols. M joins dot to dot and square to square; R, after a half-turn, joins dot to square. Use a small joining tab or external tape without covering a cut route. Adults cut and prepare all pieces; the student diagrams are recipes, not actual-size cutting templates.

For six full pair-sized kits across K–1 / 2–3 / 4–5: 78 strips (24 lane strips,18 reusable seam strips,24 puncture strips,12 spare design strips), six tape rolls/dispensers and trays,72 edge dots. Practical minimum can be smaller: two P3 kits at K–1, two P1/P2 kits at 2–3 and two full kits at 4–5. Preserve unused objects for later visits. Preparation is substantial if every model is made simultaneously; select one investigation per visit and reuse P2 strips. Three anchored adults can support the prepared kit, but simultaneous cutting for several pairs needs advance preparation rather than in-session adult bottlenecks.

For A/B remove one radius-5 mm disk at longitudinal center, centered across the width; A has M seam and B R. C has R seam with a radius-5 mm semicircular notch in one long edge midway along the strip. D has M seam with radius-5 mm half-disks centered in both short ends; after joining they form one disk. Place tape around the remaining seam portions without filling or bridging D's hole. For four-boundary designs, use M with two interior disks spaced 50 mm apart, or R with three disks spaced 50 mm apart; keep all disk closures at least 15 mm from any old boundary/seam. No hole reaches across the width.

Launch together: handle an uncut band, choose a start dot, and follow its edge through the marked seam until first return. Show the non-task two-lane flat recipe, follow one lane into its marked

continuation, and keep “piece” distinct from “closed edge.” Do not cut this demonstration into the target results. Before P2, demonstrate the student local two-strip visual: unchanged corner marks on two short ends; M joins dot to dot; R, after a half-turn, joins dot to square; then the compact seam record. Dot still means that particular corner; an R reverses its end order by a twist, not by moving or renaming labels. This is a local join example and does not close a band or reveal the target boundary counts. Before P3 show a flat paper washer: one piece can have two edges. If a K–1 child cannot follow a real seam/edge without switching, remain with that concrete tracing rather than asking them to operate a recipe.

A return visit can explore P1 for 20–35 minutes; P2 or P3 can each fill 20–40 minutes on another visit. Finish with a saved seam record and actual edge counts. All timings, seam/cut handling, tape fit and child transfer are **unpiloted and physically untested**. An adult should rehearse each exact model before classroom use. Digital verification does not claim the rehearsal occurred.

Solutions, hints and extensions

P1 / student p. 1. 3M: three annuli, two edges each, total six. 3R: outer lanes 1/3 make one annulus, middle lane 2 remains a Möbius band; two pieces, edge counts 2 and1, total3. 4M: four annuli, total8. 4R: lane pairs1/4 and2/3, two annuli, total4. The mixed one/two-edge result occurs in 3R. All dotted lines must be cut as complete closed lane boundaries. A reversed seam may carry one printed cut line to another; follow through the seam until the complete cut curve returns, and do not stop after one traversal of the flat strip. Hint: trace a lane label through the seam before naming pieces. Extension: predict five and six lanes using paired positions; explain the single middle lane only for odd n. These are variants of the lane investigation, not extra counted investigations.

P2 / student p. 2. MMM has2 edges; MMR1; MRR2; RRR1. The position of the R seam(s) does not affect the parity. Four-strip one-edge example RMMM; two-edge example RRMM (or MMMM). Follow a transverse top/bottom marker around the chain: an R toggles it, M keeps it. Two R seams cancel the reversal; any odd total returns the marker on the opposite side and the edge trip continues along the other long edge. Hint: use a two-sided arrow at one unchanged starting seam, rather than record several changing orientations simultaneously. Extension: compare different sequences with equal parity, and explain what counts ignore about physical knotting or linking.

P3 / student p. 3. A3 edges, B2, C1, D3. All remain one connected piece. A's disk adds one to the annulus's2. B's disk adds one to the Möbius band's1. C's notch stays within the existing single edge; follow around its indentation without starting a new trip. D's matching half-holes become one hole, so the annulus has3. A connected four-edge witness is M with two small disks (2+2=4); another is R with three (1+3=4), within the stated limit. K–1 can observe the separate inner edge(s) without learning “puncture.” To explain preserved connectivity, replace any path through the removed disk interior by a detour along its retained cut rim; the removal does not cut all the way across the strip. Hint: put a removable dot on each discovered closed edge to prevent repeated counting. Extension: draw a proposed seam-crossing hole on the flat recipe and decide which pieces of its boundary must be paired; two unmatched half-holes are a different object.

Distinction and checks

Base Week38 studies one middle cut, dot grouping and transported arrows. P1 introduces three/four lane cycles and mixed cut pieces; P2 composes several separately marked seams; P3 creates boundaries with punctures and contrasts notches, including a seam-crossing hole. These original extensions reuse the concrete band action, not the base target enumeration or Week1 tilings.

student/verify.py computes all printed lane cycles, independently traces cyclic top/bottom boundary segments for every M/R sequence of lengths1–5, and counts surface/boundary components in a square-cell model of each puncture/notch recipe. The mesh uses homeomorphic square holes, not a physical cutting simulation; the small-disk argument connects it to the round printed examples. It separately checks hole width/separation and the two four-boundary witnesses. Portable source rebuilding uses sh student/build.sh OUTPUT_DIR, Python3 and ReportLab, with no repository imports. Record use as W38-B1/B2/B3 and distinguish observed constructions from explanations. All material remains unpiloted; exact physical pretests are unperformed.